Ajay Rajendra Kumar

RESEARCH ASSISTANT · VISION-LANGUAGE MODELS

Boston, MA, USA

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"Towards Generalizable and Reliable Artificial Intelligence"

Summary

Computer science graduate conducting research in computer vision and multimodal learning using multimodal large language models (MLLMs). Focused on building models that can perceive and reason over vision/audio inputs in long-horizon settings, with a particular focus on using agentic architectures to perform active perception and improve spatial and causal reasoning abilities in MLLMs. Research assistant in Prof. Lorenzo Torresani's lab at Northeastern University; previously part of the Visual Intelligence Lab at Northeastern.

Highlights

- Co-authored a workshop paper accepted at ICRA 2026 which was also accepted at CVPR 2026 (Demo Track).
- Academic honors & awards:
  - Most Innovative Project among 40+ submissions, PhD-level Deep Learning Course, Northeastern University (Mar. 2025).
  - National 2nd Runner-Up, Daimler India Commercial Vehicles Hackathon (Aug. 2022).
  - 1st Runner-Up, Transfusion Medicine Hackathon (Apr. 2022).
  - Winner, Space Odyssey Web Development Hackathon (Jun. 2021).
  - Scholar's Scholarship Award, Manipal Academy of Higher Education (Jan. 2021).
- Research experience across computer vision and NLP at Northeastern University (Prof. Lorenzo Torresani, Prof. Huaizu Jiang), IIT Kharagpur, Institute for Plasma Research, and Bhabha Atomic Research Centre.
- Industry experience as a Software Development Intern at Fidelity Investments; received a full-time return offer.

Field of Interest

MY RESEARCH INTERESTS INCLUDE, BUT ARE NOT CONFINED TO:

- Long-horizon multimodal reasoning.
- Multimodal representation learning and evidence retrieval.
- Continual learning and lifelong memory for MLLMs.
- Dense visual SLAM and 3D scene reconstruction.
- Efficient deep learning for real-time perception.

Education

Northeastern University
Boston, USA

M.S. IN COMPUTER SCIENCE — *Sep. 2024 - May 2026*

- GPA: 4.0/4.0
- Capstone: "Dense Visual SLAM on Edge Devices"
- Advised by Prof. Huaizu Jiang

Manipal Institute of Technology
Manipal, India

B.TECH. IN COMPUTER SCIENCE — *Aug. 2020 - Jul. 2024*

- GPA: 9.3/10.0
- Minor: Computational Intelligence
- Dissertation: "Evaluating Deep Learning Approaches for Fracture Segmentation in X-ray Images"
- Advised by Prof. Subhamoy Mandal

Work Experience

Research Assistant
Boston, USA

NORTHEASTERN UNIVERSITY, KHOURY COLLEGE OF COMPUTER SCIENCES — *Nov. 2025 - Present*

- Serving as a research assistant at Prof. Lorenzo Torresani's lab.
- Working on improving causal reasoning capabilities in vision–language models.

Research Assistant
Boston, USA

NORTHEASTERN UNIVERSITY, KHOURY COLLEGE OF COMPUTER SCIENCES — *Apr. 2025 - Mar. 2026*

- Served as a research assistant at Prof. Huaizu Jiang's lab.
- Worked on building a dense visual-SLAM system capable of running real-time on edge devices.
- Submitted a paper to IEEE IROS 2026 (under review).
- Demo accepted at CVPR 2026; workshop paper accepted at ICRA S2S Workshop 2026.

Research Assistant
Boston, USA

NORTHEASTERN UNIVERSITY, D'AMORE-MCKIM SCHOOL OF BUSINESS — *Feb. 2025 - May 2026*

- Served as a graduate research assistant at Prof. Anna Lamin and Valentina Marano's lab.
- Contributed to projects to evaluate CSR initiatives driving social impact and sustainable development outcomes in India.
- Collaborated with Indian Institute of Management Bangalore on analyzing the impact of corporate growth in Bengaluru on housing.

Undergraduate Researcher (Bachelor's Thesis)
Kharagpur, India

INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR — *Jan. 2024 - Jul. 2024*

- Advised by Prof. Subhamoy Mandal.
- Worked on semantic segmentation of humerus bone from X-ray images.

Software Development Intern
Bengaluru, India

FIDELITY INVESTMENTS — *Jun. 2023 - Aug. 2023*

- Redesigned change ticket management system using Angular and Spring Boot.
- Obtained a full-time offer to join Fidelity.

Project Intern
Gandhinagar, India

INSTITUTE FOR PLASMA RESEARCH — *Apr. 2023 - Dec. 2023*

- Advised by Prof. Rajaraman Ganesh.
- Worked on simulation of turbulent fluid flows.
- Delivered experimental simulation software to the computing division of IPR.

Project Intern
Visakhapatnam, India

BHABHA ATOMIC RESEARCH CENTRE — *Dec. 2022 - Jan. 2023*

- Advised by Dr. Manoj Warrier.
- Contributed to early-stage research on modeling spectral neighbor analysis potentials for zirconium using machine learning techniques.

Selected Projects

Pose Graph Sparsification by Maximizing Algebraic Connectivity
Northeastern University, Boston, USA

MOBILE ROBOTICS — *Mar. 2025*

- Formulated and implemented an iterative variant of the algebraic connectivity maximization algorithm for pose graph sparsification.
- Evaluated on benchmark pose graphs using GTSAM and SE-Sync, achieving sparser graphs with minimal performance loss.
- Keywords: Pose Graph Optimization, Graph Sparsification, SLAM, GTSAM, SE-Sync

Hallucination Detection in Large Language Models
Northeastern University, Boston, USA

NATURAL LANGUAGE PROCESSING — *Mar. 2025*

- Evaluated hallucination detection methods from "Joint Evaluation of Answer and Reasoning Consistency for Hallucination Detection in Large Reasoning Models" (AAAI 2026) across multiple open-source LLMs.
- Identified limited generalization of the proposed methods beyond the original evaluated models through thorough experiments.
- Keywords: Hallucination Detection, Large Language Models, PyTorch, Python

Improving Text-Driven Human Motion Generation through Equivariance
Northeastern University, Boston, USA

DEEP LEARNING — *Mar. 2025*

- Integrated equivariant neural networks into motion diffusion models for human motion generation.
- Improved matching accuracy and achieved competitive FID scores against existing baselines.
- Keywords: Human Motion Generation, Diffusion Models, Equivariant Neural Networks, Motion Diffusion, PyTorch

Rethinking Depth for Object Detection with YOLOv3
Northeastern University, Boston, USA

COMPUTER VISION — *Dec. 2024*

- Implemented YOLOv3 and RGB-D variants to study the impact of depth cues on object detection.
- Observed limited performance gains, indicating need for improved depth fusion architectures.
- Keywords: Object Detection, RGB-D Vision, Depth Fusion, YOLOv3, Multimodal Perception, OpenCV, Python

Addressing Vaccine Misinformation on Social Media
Manipal Institute of Technology, Manipal, India

RESEARCH ASSISTANT (BACHELOR'S) — *Mar. 2024*

- Fine-tuned transformer models for misinformation classification on social media data.
- Developed similarity-based method for anomalous user detection, achieving strong classification performance.
- Keywords: Misinformation Detection, Social Media Analysis, Deep Learning, PyTorch

Publications

Underline denotes co-first authors or co-corresponding authors

International Conference

1. Aniket Gupta, Tianye Ding, **Ajay Rajendra Kumar**, Dennis Giaya, Pia Bideau, Charles Saunders, Qu Cao, Aruni RoyChowdhury, Hanumant Singh, and Huaizu Jiang, “NeuSLAM: Dense Visual SLAM on Edge Devices,”; accepted to *ICRA 2026 Workshop on From Sea to Space: Advancing Perception in Harsh Domains*; accepted to *CVPR 2026 Demo Track*.
2. C. Rao*, G. M. Prabhu*, **Ajay Rajendra Kumar***, S. Gupta, and N. P. Shetty, “Addressing Vaccine Misinformation on Social Media by Leveraging Transformers and User Association Dynamics,” in *Int. Conf. Mach. Learn. Data Eng. (ICMLDE)*, 2023.

Teaching Experience

Object-Oriented Design (CS2163) — *Manipal Institute of Technology, Manipal, India*

Undergraduate Teaching Assistant (Instructor: Prof. Anup Bhat) — *Aug. 2023 - Dec. 2023*

Problem Solving Using Computers (CS1081) — *Manipal Institute of Technology, Manipal, India*

Undergraduate Teaching Assistant (Instructor: Prof. T. Sujithra) — *Aug. 2023 - Dec. 2023*

Honors & Awards

Mar. 2025	**Most Innovative Project (among 40+ submissions)**, PhD-level Deep Learning Course (CS7150), Northeastern University	*Boston, USA*
Oct. 2023	**Undergraduate Sponsorship Grant**, Manipal Academy of Higher Education (MAHE)	*Dehradun, India*
Aug. 2022	**National 2nd Runner-Up**, Daimler India Commercial Vehicles Hackathon	*India*
Apr. 2022	**1st Runner-Up**, Transfusion Medicine Hackathon	*Manipal, India*
Jun. 2021	**Winner**, Space Odyssey Web Development Hackathon	*Manipal, India*
Jan. 2021	**Scholar’s Scholarship Award**, Manipal Academy of Higher Education (MAHE)	*Manipal, India*